

Member ID: _____

Time: _____

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LINUX OPERATING SYSTEM FUNDAMENTALS

(350)

STATE 2023

Total Points: (50 @ 2 points each) _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. The Linux utility `gfdisk` can be used to partition and format disks using new GPT partition table?
 - A. True
 - B. False
2. In Ubuntu when formatting a partition with `mkfs` if you don't specify a filesystem what default filesystem will it format as?
 - A. XFS
 - B. Ext2
 - C. Ext3
 - D. Ext4
3. Which of the commands below would not be a valid command to list partitions of disk `/dev/sdb`?
 - A. `fdisk -l /dev/sdb`
 - B. `cgdisk -l /dev/sdb`
 - C. `gdisk -l /dev/sdb`
 - D. `parted -l /dev/sdb`
4. Looking at the output below which is a hidden file?


```
drwxrwxr-x 2 ryan ryan 4.0K Aug 14 19:57 test
drwxrwxr-x 2 ryan ryan 4.0K Aug 14 20:06 .test1
-rw-rw-r-- 1 ryan ryan 0 Aug 14 19:57 .test2
-rwxrwxr-t 1 ryan ryan 0 Aug 14 19:58 test3
```

 - A. test
 - B. test1
 - C. test2
 - D. test3
5. Which command below is the correct way to set a persistent hostname of "BPA-Server" in Ubuntu?
 - A. `sudo systemctl set-hostname BPA-Server`
 - B. `sudo hostnamectl set-hostname BPA-Server`
 - C. `sudo cat BPA-Server > /etc/hostname`
 - D. `sudo hostname --set BPA-Server`
6. Adding the argument "`-t`" to the command `ls` will sort the output from newest to oldest?
 - A. True
 - B. False

7. Which chmod command(s) below will make the file setup.sh executable?
 - A. chmod +X ./setup.sh
 - B. chmod +x ./setup.sh
 - C. chmod 777 ./setup.sh
 - D. B and C

8. The command “**sudo mount /dev/sda2 /Data**” will successfully mount the second partition of the first disk to the **/Data** directory.
 - A. True
 - B. False

9. Which command below will successfully list the size and free space of a volume group named Shared?
 - A. sudo lvm Shared
 - B. sudo lvs Shared
 - C. sudo pvs Shared
 - D. sudo vgs Shared

10. There is an LVM Volume group named Data on this system and you’ve been asked to create a new logical volume in that volume group named Shared. That logical volume needs to be 6GB in size. Which command below will successfully complete that task?
 - A. sudo lvcreate -n Shared Data -l6GB
 - B. sudo lvcreate -n Shared --vg Data -l6GB
 - C. sudo lvcreate -n Shared Data -L6GB
 - D. sudo lvcreate -n Shared --vg Data -L6GB

11. Which command(s) below will display the ipv4 address of all network adapters on the system omitting ipv6 addresses?
 - A. ip -4 a
 - B. ip -v4 a
 - C. ip -4 address
 - D. A and C

12. Which command below will successfully **change** the IP Address of the Network Adapter **em1** to **192.168.2.5** with subnet of **255.255.255.0**?
 - A. sudo nmcli connection modify em1 ipv4.addresses 192.168.2.5/255.255.255.0
 - B. sudo nmcli connection modify em1 ipv4.addresses 192.168.2.5/24
 - C. sudo nmcli connection add em1 ipv4.addresses 192.168.2.5/255.255.255.0
 - D. sudo nmcli connection add em1 ipv4.addresses 192.168.2.5/24

13. A partition spanning the full disk has been created on disk **sdb** for use with LVM. What command below would correctly create a volume group named Shared using that partition in Ubuntu?
 - A. sudo vgcreate -n Shared -d /dev/sdb1
 - B. sudo vgcreate Shared /dev/sdb1
 - C. sudo vgcreate -n Shared -p /dev/sdb1
 - D. sudo vgcreate Shared sdb1

14. The command “scp” can be used to transfer files over the network via the ssh protocol.
- A. True
 - B. False
15. FirewallD is the default firewall used in Redhat Linux version 7 and newer.
- A. True
 - B. False
16. Which command below would correctly set the NIC ens192 to use Static assigned IP instead of DHCP?
- A. sudo nmcli connection edit ens192 ipv4.method manual
 - B. sudo nmcli connection modify ens192 ipv4.method static
 - C. sudo nmcli connection edit ens192 ipv4.method static
 - D. sudo nmcli connection modify ens192 ipv4.method manual
17. Which command below would correctly change the GID of a user named Rob to 1007?
- A. sudo usermod -G 1007 Rob
 - B. sudo usermod -g 1007 Rob
 - C. sudo useradd -g 1007 Rob
 - D. sudo useradd -G 1007 Rob
18. What Linux command below would correctly create a new user called John with a UID of 1004?
- A. sudo useradd -U 1004 John
 - B. sudo usermod -u 1004 John
 - C. sudo useradd -u 1004 John
 - D. sudo usermod -U 1004 John
19. On the server is a user named **Paul** and he needs to be added to the two groups, **Research** and **Development**, without affecting his primary group. Which command below would successfully accomplish this?
- A. sudo usermod -g Research, Development Paul
 - B. sudo usermod -aG Research,Development Paul
 - C. sudo usermod -ag Research, Development Paul
 - D. sudo usermod -G Research, Development Paul
20. You need to add a new user named **Larry** but they need their home folder to be **/mnt/Users/Larry** instead of in the default /home directory, which command below would successfully accomplish this?
- A. sudo useradd -h /mnt/Users/Larry Larry
 - B. sudo useradd -d /mnt/Users/Larry Larry
 - C. sudo useradd -m /mnt/Users/Larry Larry
 - D. sudo useradd -s /mnt/Users/Larry Larry

21. The command “**vgcreate**” is used to format a disk or partition for use with LVM volume groups.
- A. True
 - B. False
22. The Linux command ‘**lsblk**’ will list all disks and their partitions?
- A. True
 - B. False
23. In Ubuntu to allow a user to use the sudo command they must first be added to the root group.
- A. True
 - B. False
24. What command below will update the available packages in Ubuntu but not install them?
- A. sudo apt update
 - B. sudo dpkg --update
 - C. sudo apt -l update
 - D. sudo apt -t upgrade
25. The minimal install of Ubuntu Server will include a default Unity GUI desktop.
- A. True
 - B. False
26. The Ubuntu command ‘**apt search ssh**’ would search repositories for any packages matching the name **ssh**.
- A. True
 - B. False
27. Which command below can be used to upgrade Ubuntu 20.04 to Ubuntu 20.10?
- A. sudo apt dist-upgrade
 - B. sudo do-release-upgrade
 - C. sudo apt upgrade
 - D. sudo apt full-upgrade
28. In Ubuntu which command would set the SSH service to start on boot?
- A. sudo systemctl start sshd
 - B. sudo systemctl enable sshd
 - C. sudo systemctl enable --now sshd
 - D. sudo systemctl is-enabled --true sshd
29. Which command would stop the Samba(smb) service in Ubuntu?
- A. sudo systemctl stop smb
 - B. sudo systemctl disable smb
 - C. sudo systemctl stop --disable smb
 - D. sudo systemctl is-disabled --now smb

30. Using the journalctl utility to view system logs what command below would allow you to filter the logs for anything referencing the **ssh** service?
- A. `sudo journalctl -g ssh`
 - B. `sudo journalctl -u ssh`
 - C. `sudo journalctl -t ssh`
 - D. `sudo journalctl -S ssh`
31. Which command below will filter journalctl log output showing only warning and above type entries?
- E. `sudo journalctl -s warning`
 - F. `sudo journalctl -p warning`
 - G. `sudo journalctl -f warning`
 - H. `sudo journalctl -b warning`
32. Looking at the attributes below for the file labeled “file1” you can conclude that anyone in the group admins would have both read and write access to the file.
- rw-r--rw- 1 ryan admins 0 Aug 28 18:08 file1**
- A. True
 - B. False
33. Looking at the attributes below for the file labeled “file2” who would have read and execute access but not write access?
- r-xrw---x 1 ryan zack 0 Aug 28 18:11 file2**
- A. ryan
 - B. zack
 - C. Everyone
 - D. Neither
34. You have a user that says they can’t open a folder called **/Project** and get permission denied when trying to run the command “`cd /Project`”. You have already confirmed their group has been assigned to the folder. What directory permissions do they need added for their group in order to open the folder and view the contents of the folder?
- A. Write
 - B. Read
 - C. Execute
 - D. B and C
35. If you were to change permissions to a file using the command: “`chmod 766 filename`” what would the resulting permissions look like when viewing the file?
- A. `-rwxrw-rw-`
 - B. `-rwxrw----`
 - C. `-rwxrwxrwx`
 - D. `----rwxrwx`

36. Review the permissions below for the folder Database. You've been asked to change permissions so that only the members of the Management group have read-only access. Which command(s) would accomplish this?

drwxr-xrwx 2 root Management 4.0K Aug 28 18:34 Database

- A. `sudo chown 767 Database`
 - B. `sudo chmod o-w Database`
 - C. `sudo chmod g-w Database`
 - D. `sudo chown 776 Database`
37. If you were to set permissions of **746** on a script called *runme.sh* users in which category would be given the ability to execute the script?
- A. Owner
 - B. Group
 - C. Others
 - D. None of the above
38. The `set-facl` command allows you to set permissions for users and groups on a folder or file in Ubuntu.
- A. True
 - B. False
39. Which command below would successfully filter out any mention of "error" from the */var/log/syslog* file and output to a file called */var/log/errors.log* without overwriting any existing data?
- A. `grep -i "error" /var/log/syslog > /var/log/errors.log`
 - B. `find /var/log/syslog -name "error" >> /var/log/errors.log`
 - C. `grep -i "error" /var/log/syslog >> /var/log/errors.log`
 - D. `find /var/log/syslog -name "error" > /var/log/errors.log`
40. The SSH service uses port 22 by default.
- A. True
 - B. False
41. Which firewall port would you need to enable for http on Linux?
- A. 80
 - B. 445
 - C. 443
 - D. 88
42. When viewing processes with the `top` utility which key would sort the list by CPU usage?
- A. T
 - B. C
 - C. P
 - D. M

43. Which command below can be used to only list process usage by a specific user named John?
- A. top -p John
 - B. top -S John
 - C. top -u John
 - D. top -s John
44. You have been asked to expand a volume group size by adding a new disk partition, **sdb1**, into your existing volume group named **Shared**. Which command below will accomplish this?
- A. sudo vgreszie Shared /dev/sdb1
 - B. sudo vgexpand Shared /dev/sdb1
 - C. sudo vgetend Shared /dev/sdb1
 - D. sudo vgadd Shared /dev/sdb1
45. You have been asked to remove a disk partition, sdsb1, from the volume group named Shared. What command below will successfully complete the removal?
- A. vgremove Shared /dev/sdb1
 - B. vgreduce Shared /dev/sdb1
 - C. pvremove Shared /dev/sdb1
 - D. pvmove Shared /dev/sdb1
46. The command **pvremove** is used to migrate volume group data off of a disk/partition so it can be removed from a volume group.
- A. True
 - B. False
47. Using the Linux utility nmcli you can modify network settings and set NICs to start up at boot.
- A. True
 - B. False
48. Which command below would be used to create a new grub config on Redhat Linux?
- A. grub2-install
 - B. update-grub
 - C. grub2-mkconfig
 - D. update-grubconfig
49. In Redhat Linux selinux is disabled by default.
- A. True
 - B. False
50. The Samba service provides file and print services for both Linux and Windows client systems.
- A. True
 - B. False